Philips HealthSuite Digital Platform: Week 1 Case Study Summary

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I. Introduction

The healthcare system is undergoing a critical change, shifting from localized data storage and processing to a highly integrated cloud-native platform. Philips HealthSuite Digital Platform (HSDP) is at the forefront of this change. It is a Platform-as-a-Service (PaaS) built natively on Amazon Web Services (AWS) that merges fragmented clinical records, medical imaging, and consumer wearable records within a regulatory-compliant cloud environment [1], [2]. This report discusses the problems of on-premises management and the salient features of HSDP.

II. On-Premises Management Problems and Bottlenecks

Globally, almost 3.6 billion medical imaging procedures are performed annually across different healthcare organizations [3]. Within the United States, data indicates that the average workload assigned to a radiologist nearly doubled between 2008 and 2023, rising from 58 imaging procedures per day to 100 per day [3]. Simultaneously, higher-resolution images caused the average file size of a single imaging study to double to nearly 150 MB [3]. Traditional on-premises infrastructure for storing them consisted of picture archiving and communication systems (PACS) and Vendor Neutral Archives (VNA). The rapid increase in the frequency as well as the size of these images overwhelmed these storage options. This meant requiring a dedicated team of IT personnel to maintain the servers, procuring hardware, and dedicating space to physically house the servers. For data availability, multiple copies were generally maintained, which meant more storage cost and the overhead of maintaining consistency [3].

Furthermore, these institutions required separate sets of software to access different kinds of information, such as reports, image archives, and patient records, because each of them was proprietary software with little flexibility [3]. This is basically known as data fragmentation.

Another overhead was making sure that all the stored data was compliant with legal requirements such as the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in Europe [2].

III. Migration from On-Premises to Cloud Native

Philips' cloud migration can be classified into two phases. First, it migrated onto AWS using core AWS compute and storage services [4]. Second, it changed how it utilized AWS. When HSDP launched in 2014, Philips built and ran many of its services on top of core AWS infrastructure, such as container hosting, notifications, and medical image (DICOM) and clinical data (CDR) stores [5], [6]. With the passage of time, AWS released its own services focused on the healthcare sector, such as AWS HealthImaging [3]. Philips began replacing its proprietary solutions in favor of AWS-provided solutions. CDR and DICOM services were removed as end-of-life in October 2025, and Container Host, Docker, and PKI were dropped from Philips' infrastructure tooling in June 2026 [6]. Philips now reports using more than 100 AWS services [4]. Some legacy services, such as Cloud Foundry, are still listed as running [7], so this is a gradual shift rather than a completed switch.

IV. Architecture of HSDP

HSDP is an abstraction layer that allows healthcare to become cloud native [2]. The architecture is divided into several domains: Connect, Host, Authorize, Store, Analyze, and Share [8].

i) Connect: It is used for secure and high-speed ingestion of data from a variety of devices, such as medical imaging instruments, consumer wearables, and patient monitoring devices. It is also used for pushing software and firmware updates to these devices [2]. Philips reported that over 1.3 million devices are connected to this service [4]. At its core, it uses AWS IoT Core to manage millions of IoT devices and route trillions of messages securely [9].

ii) Authorize: It provides Identity and Access Management across multiple services and consumers, along with data compliance [2].

iii) Host: It provides the application execution environment, databases, message queues, and centralized logging, while ensuring compliance with high availability [2]. At its core, HSDP relies on Amazon Elastic Compute Cloud (EC2) for computation needs [9].

iv) Store and Share: It ensures data acquisition, normalization, consistency, and availability of medical data [2]. Philips intends to use AWS HealthImaging to seamlessly manage the storage, analysis, and sharing of petabyte-scale imaging data in the cloud [3].

v) Analyze: Philips' machine learning work on AWS includes AI ToolSuite, a platform for developing ML models that is built on Amazon SageMaker [5]. Its data-labeling solution,

used to prepare training data for imaging AI, is centered on SageMaker Ground Truth [10]. Philips and AWS also built a machine learning tool that automatically finds and conceals protected health information (PHI) in medical records, so that the data can be used for model training [11]. In November 2024, Philips also announced that it aims to build generative AI applications on Amazon Bedrock, for example, conversational reporting that turns a clinician's spoken findings into a structured report for review [12].

V. Key Numbers Reported

TABLE I
KEY QUANTITATIVE OUTCOMES OF PHILIPS HEALTHSUITE ON AWS

Metric	Reported value	Relevance to our exercise	Ref.
Patient data stored on HSDP	15 PB (December 2015)	Shows the platform's scale from its early years	[13]
Connected IoT devices	Over 1.3 million (2025)	Scaling input for our cost projection	[4]
Infrastructure cost reduction	Up to 36% after on-premises data-centre migrations	Baseline for our cloud vs on-premises cost comparison	[4]
Near-real-time medical image analysis	Under 5 seconds using AWS IoT and Amazon EKS	Reference point for our alert-latency measurement	[9]
Infrastructure cost vs on-premises (same prototype)	30× lower	Supports our managed vs self-hosted comparison	[9]
De-identification of patient records	67% reduction in processing time using ML	Motivates the de-identification step in our compliance layer	[9]

Note: All figures are vendor-reported by Philips and AWS and have not been independently audited. The 5-second and 30× figures come from a prototype project, not the full HSDP platform. The 5-second figure measures image analysis time and is not directly comparable to our vitals-to-alert latency.

VI. Criticisms

Philips' own documents point to two limitations of the platform. First, vendor lock-in: HSDP is a proprietary platform, and its subscription terms allow Philips to change or remove platform features at any time, reject client applications at its discretion, and do not require it to return customer data when a contract ends [14]. Applications built on HSDP's APIs are therefore costly to move to another provider. Second, dependence on network connectivity: Philips' patient monitoring service terms require a continuous 24×7 connection to HealthSuite for certain services, and warn that a network outage can stop the central monitoring station from receiving real-time alarms from bedside monitors [15].

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